

ActInf Textbook Group ~ Cohort 4 ~ Meeting 6 (Chapter 2 part 2)

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all right hello everyone it is
July 11th and we're in the second
discussion on chapter two
so before we go into any of the
questions
does anyone just want to bring up
anything or just start with anything
about chapter two
um Ali and then anyone else who raised
their hands
uh thank you yeah I just uploaded uh the
equation to 0.5 walkthrough uh in the
um I guess I put it in the equations
table if I'm not mistaken but equation
2.6 needs a little bit more fine-tuning
before it's ready but uh I'm sure it'll
be ready for the next week so I think I
put it at the top of the
notes section of equation 2.5 so if
anyone wants to
uh
but yeah that's it
foreign
looks awesome can we maybe uh go over it
or can you just walk through it or yeah
sure yeah uh well first of all
um
we began with
um just basic definitions uh such as
some fundamental definitions such as the
Shannon entropy and then to derive uh
the first line we put we substitute the
Shannon entropy term uh into the uh I
mean the first line of equations so uh
and we can see there's uh an interesting

parallel between uh this kind of
formulation between energy and entropy
and also the path integral formulation
and variation of preology but the
essential difference here is that
here the first term represents only the
energy but for the path integral it can
be best described as the energy
constraint uh so that's
um more of a side note there and then
for the second line of equation 2.5
we get but before going into
um

the second line just let me unpack the
first line a little bit more so for for
the people who who are not familiar with
a callback Library Divergence uh we put
the definition of callback library at
the top of the
um

or yeah at the top of the page one and
then uh by using uh James inequality we
get uh the Callback the lighter
Divergence bounded by the expectation
of the $\log p$ over q and that allows us
to substitute the Shannon entropy term
into the expectation of the $\log p$ over Q
so that was the essential move for
unpacking the first line of equations so
I mean without Jensen's inequality we
wouldn't

exactly get the the exact term of the
first line of the equation because
obviously the order of the log operator
and the expectation operator is
different than uh we need here
so yeah that's basically a
straightforward algebraic manipulation

um

and the the second line of equation 2.5

or you know the terms the trade-off

between complexity and accuracy

similarly can result from

uh substituting uh I mean by expanding

uh the first line of the equation and

then substituting some of the term as

the Callback Library Divergence and then

we get the complexity term uh and the

rest will be the accuracy so this what

it means is that as pointed at the

bottom of the page the greater

complexity is uh in other words the more

one needs to change beliefs to explain

observations the Lesser predictive

accuracy will become so in this case

variational free energy uh would be just

the minimization of the complexity or at

the same time maximizing the accuracy

and again it can be

compared to the path integral

formulation it's almost identical to the

path integral formulation of the

trade-off between complexity and the

accuracy

uh all right so for this for the third

line of equation 2.5

uh we try to somehow get the trade-off

between the Divergence and the evidence

but in this case evidence is just uh

something that acts as

um as the bound or the uh evidence lower

bound uh so it's something that's always

greater than or equal to surprise right

so

by unpacking the

um the last line of equation 2.5 into

these two distinct terms or the this
trade-off between Divergence and
evidence uh we are allowed to have this
um elbow or evidence lower bound or in
other words the negative upper bound
that would enable us to regard BFE as
the optimization parameter uh or over
the parameter to be optimized rather
than uh trying to achieve the exact
value of uh free energy but just some
optimization parameter which would be
much more uh

tractable than trying to achieve the
exact amount of the free image

foreign

work

nice

cool stuff

anyone want to ask a question or or make
a comment on this

could you maybe explain a little more
like zoomed out from this document what

does the variational free energy uh
do in the model like in the textbook

yeah so A variation free energy is

a variational technique that sorry

as I mentioned allows us to do the

approximate computations of Bayesian

inference uh in terms of optimizing a

certain parameter namely the variational

free energy because Computing the exact

Bayesian inference of the orbasian

output sorry Bayesian inference

with regard to both posterior and prior

would be

almost impossible

especially in complicated complicated

situations and by complicated I mean

everything that's not trivialism a simple situation so that would basically uh are all real-time situations that we encounter um and uh obviously we're interested in modeling so this this approximate Bayesian inference technique is a very useful way to overcome this shortcoming uh exact Bayesian inference in terms of its of its intractability and it makes the computation of or the optimization of the variation of free energy much more computable and tractable as compared to um exact Bayesian inference uh which would obviously have uh many practical um

I mean it can have uh many practical benefits uh and not just something uh that's described in theory and doesn't have any practical applications and modeling and so on so yeah that's uh basically the idea behind using the variational technique and using a certain parameter as the upper bound or in other words the negative of a lower bound

that needs to be optimized whether maximize or minimized um instead of just trying to compute the exact value of Interest

so variation free energy on the other hand

includes all the

um

established Frameworks for doing Bayesian inferences so for example the techniques developed by James Kahneman

and others that I I think it was

um

which table was um which figure I in one

of the figures in the book I I suppose

it was in chapter two I guess uh there's

a nice comparison between uh the

different terms of variation of free

energy and all the other theories of uh

Bayesian inference yeah exactly

so we can see that it's

um more generalized theory of optimizing

the Bayesian inference than risk

sensitive kale control and other

techniques of amazing inference that's

that have been developed throughout I

mean the past several decades and um

so

each of those terms

can be

accounted for

by including or let me put it this way

by vomiting some of the terms and we can

basically get the exact same formulation

or the exact same previous formulations

we had before but as we see this is the

much more uh General way of

um

calculating the approximate Bayesian

inference

okay

Anuj do you want to ask your question or

do you want me to read it

yeah I just had this curious like uh is

that case like when we already know that

uh like I cannot do exact inference that

are there cases when uh I should also

avoid optimizing this variation free

energy given that in any case I cannot

do exact movement

I'll give a first thought on that you

always could you could choose your

models however you want but you will be

doing strictly worse with respect to the

accuracy complexity trade-off

so you could but you can invent an

arbitrary statistical method

but you will not outperform the um

information geometric

maximum possible

um performance

yeah thank you

in any given free energy that you do

compute

the the model may just be rolling to the

bottom of the hill and you might just be

way off base for a real system

so this isn't the model adequacy

problem

this is the model

optimization problem

so it's analogous to the l-tune norm in

the linear regression framework like

every scatter plot you always converge

to the one L2 Norm

progression r squared p-value all of

this

so this plays a somewhat of an analogous

role

um by enabling these uh different uh

generative models

to be optimized in a real-time

sequential way

and it turns out that it's the same

um optimization strategy that

variational autoencoders use

uh and just variational methods in

physics

so it's not like variational methods

comes from fep or or active

it's just using Bayes optimal model

fitting

just like L2 regression let's just say

is optimal linear regression fitting

optimal base fitting with the

variational free energy

which is a functional

because it's a function of a function

it's a function of Q the function that

you're optimizing over and why the data

that are coming in

so generative model of vision and

sensory incoming data which is what is

like the thing that is is getting

the best trade-off

of accuracy and variance

thank you

okay and then

um

Ali or oh and yeah go for it

uh sorry yeah and one

um I think crucial Point here is uh the

reason for using

um the Callback Library Divergence for

comparing uh the two distributions so uh

I suppose the justification for using KL

Divergence is not explicitly

stated in the textbook but it it

derives from the name in Pearson Lemma

which is basically says something to the

effect that most efficient way to

compare two distributions would be to

compare the logarithm of their I mean to

compute the logarithm of their

likelihood ratios so that's basically

what's meant by callback libler so it's
not just an arbitrary technique for
comparing to this distributions it's
actually a proven theorem or Lemma that
that this is actually actually the most
efficient way to compare two
distributions so uh

yeah I wanted to point that out as well
yeah thanks and just adding one piece
I'll put this in the notes there

this is uh with nor Sajid and others and
Carl

and they use the Rainy Divergence so in
principle

you could use different Divergence or
distance measures

but

the kind of um I mean it's a it's
definitely an interesting question about
where one might have different
properties but KL is kind of like
the go-to in at least this sense

yeah Alexi and then anyone else

this may seem like a silly question I
think Ali gave a fantastic mathematical
uh Road uh but is there a metaphor for
variational free energy versus free
energy that's free energies explained in
the book well the variational cell this
one is intractable this one is tractable
this is an estimation but

uh I don't know in terms of you know uh
is there an example or a metaphor that
can bring it home a variational for
energy specifically does anybody have a
yeah I'll think about it I'm happy to
say one but does anyone else want to
you have a thought like

what is free energy and you know and
variational free energy
how can we think about these
Ali that anyone else who wants to
um yeah I'm not sure about
um particular metaphor but uh you see uh
the I mean the optimization approach as
opposed to calculating the exact value
of something that basically comes
uh or it can be uh likened to the
situation we have uh when uh when very
in a very simple situations uh we have a
function uh that can get an exact value
of something as compared to the
asymptotes of uh getting increasingly uh
close to the to the value but never
reaching the value so uh by optimization
we allow for even uh the exact even for
the situations for for which we don't
have any information uh with which we
would be able to calculate the exact
value of something but instead we can
get closer and closer to the exact value
with the information we have but how but
nonetheless we can reach a very good
approximation and obviously we can
Define uh what is what do we mean by the
good approximation based on uh the error
we we want to reach but uh yeah in most
situations we don't have enough
information to calculate the exact value
of the parameter we want to calculate so
it's more than the question of
intractability versus tractability in
some cases it is uh I mean impossible to
compute the exact value based on the
information we have so
um I believe in these cases the

optimization technique or variational
technique is the only way we can get
close enough to the real value
or to the exact value
that's great

Point

um that like even if the the variational
model is just woefully not structurally
mapping onto the real
external uh generating function
like the generative model is a unimodal
distribution

um and then the the out there in the
world there's like five bins
and so then there would be different
patterns of model fit like it might just
fit the one that has the most or the one
not the most or blur the whole thing but
according to its generative model
it would never get worse in terms of
all of the considerations in equation
2.5

it will never get worse on the
complexity minus accuracy Frontier or it
will learn that trade-off manifold
optimally
so those are some fairly
um

good statistical uh aspects just of any
Bayesian model that's fit this way
so that doesn't guarantee like adequacy
in the real world because again you
could just have an inadequate generative
model then cease to get to the charging
station or whatever

um so it's not a success or a function
guarantee but it's a model fit
or it doesn't mean that you'll have um

adequately balanced

you know or had a memory window that's optimal or all these other things unless you've really taken them all into account

and and then just to Alexi's question uh or Broadway do you want to add anything um

equation 2.6 is the expected for energy I'll leave maybe do you want to summarize how is equation 2.5 different than 2.6

Yeah so basically

um they're very very similar structurally uh but uh obviously we need to take into account uh the pi or the policy that needs to be

um I mean in variation free energy we we only deal with the situation in which uh the perceptual data is um we only deal with perceptual data so we don't need to uh

act upon the environment in order to fit the data better to our generative model but in expected free energy

we delve into the action domain of the I mean action perception so obviously in this case we need to take into account uh the effect of the policies on our both priors and posteriors so uh the point of commencement uh for for

derivation of equation 2.6 from equation to 0.5 is exactly how applying the policies to those prior and posterior terms can affect uh the whole equation and that's where we get equation 2.6 and its uh variance in other lines of the equation

awesome

one point about equation 2.6 is uh first
off it's G not f

and it is uh like the argument that it
takes is π which is the vector of the
list of all of the policy trajectory
possibilities

so variational free energy which was
described

um really comprehensively earlier is
basically that trade-off Frontier
um between accuracy and variance
estimator

according to a generative model that's
the Q and the Y on the variational free
energy this is like a sense making or
perceptual real time and the
self-evidencing component of variational
free energy is like if you're repeatedly
measuring your body to be at homeostasis
and you expect and prefer that then like
things are working out and that's sort
of the closure

of the adoptive organism is like by
fitting the variational free energy
about homeostasis

then it stays within homeostatic bound
but expected free energy uses this π
or policy variable

and so

whereas variational free energy is more
like signals processing expected free
energy is more like control theory
because it's an explicit evaluation
of

different possible futures

does anyone have any other like thoughts
or or ways of thinking about

variational and expected free energy

um

I did have a this is more of a form of question sorry but uh I thought it might be useful so

we have here in chapter two page 34.

um when evaluating the free energy of outcomes the outcomes are the consequences however when evaluating the expected free energy the outcomes play the role of causes and since they're variables that are in the future but explain decisions of the present

um would you say that everything you said this far is kind of

gotten to that

um kind of explaining what that quote means

maybe paste the quote here just so we can see it

thank you meanwhile

let's let's uh look at this

question can anyone throw some light on these terms both are posterior so indeed

sometimes like the like initial prior D is called like the prior but technically every variable that gets an observation coming to it in a Bayesian framework is a prior an observation in a posterior update on something it's not like a

free-floating prior so every

distribution that's Bayesian

that's why it's like prior on state

space

or prior on policy which is like habit

that gets sharpened by expected free energy so then the output of this

operation is like

habit

the policy prior sharpened by g into

policy posterior

and then that policy posterior can be

sampled from or just selected with the

best option

um then about the μ specific

interests you paste it in all μ we'll

look at it but just now to these two

terms mentioned so $Q(x)$ given p_i and $Q(x)$

given Y and P_i so this is a question

about the KL Divergence

um so

the the

computation of the expected free energy

for a given policy

everything is conditioned on like that

policy happening and so between these

two things uh two sides of the double

line

they're both conditional policy and

they're both have an aboutness of the

Hidden State Through Time X_{t-1} so the

only difference is that there's that

there's this y_t observations

Through Time on the left side so if the

observations that you expect under that

course of action

aren't going to update your hidden State

beliefs

then the the Divergence between these

two cues is zero the information is

meaningless there's no information

gained

you're not learning anything you're not

updating your hidden State model

based upon the observation whereas if it

was an epistemically informative course

of action and do you expect it to reduce
your uncertainty

um on this course of action

then y tilde

would lead to an update on X tilde so

there would be a Divergence here

so that policy would have epistemic
value

whereas the pragmatic value

is about the alignment of preferences
with the observations that you expect

all right

does anyone have a thought on or or on

on this quote from Andrew

I'll just give one reading not sure if

this is the the full um or only way

which is like

um

when we're talking about the uh the

sensory data themselves which are some
sometimes just called outcomes sensory
outcomes

so like when we're talking about

the the why that were

um the outcomes that we're actually
observing

um those are the consequences of hidden
states that are like emitting those

so we're doing we're doing equation 2.5

um where the consequences of the causal
model

are what we're trying to fit

um in contrast in 2.6 those observations

in y tilde have not happened yet

so they're projected

observations

so they are um unobserved like it's not

it's the opposite of time travel say you

and I haven't observed the temperature
in two hours

but I can have a tight or or a wide uh
prior on is that variable but it's um
it's a type of hidden cause analysis
to say well if this happens under this
condition

and then basically kind of back
propagate that into the present
so as a causal impact of an
informational State that's expected
and that's not like a paradox or
anything like that

but so there are the outcomes of a
causal process but then in the efe
they also have a causal
Force because differences in them lead
to differences different differences in
expected observations lead to different
action selection

anyone else want to ask a question or it
looks like there's at least one that's
from the cohort

Alexi yeah

um so I was really uh
happy to to read this paragraph where
they talk about generative process and
generated model and they're saying that
uh they could be in different state
spaces one could be five dimensional one
could be two-dimensional continuous and
categorical I think that's a very
profound statement uh and
uh for example if I talk about
temperature outside and I use words
and I see things like hot and cold right
so I I place a continuous
outside variable in my two categories so

I could model this way but it's it's a
bit of a distortion if you wish
but I don't know if you have any
comments I'm just thinking out loud
about this that
this this is this is a thing it's just
mind-blowing that you know we're
modeling something but we will be in a
completely different state space and
different variables different variable
structure right
yeah that's a great Insight uh one thing
I'd add to that is like
um if you measure the the
height of the people in the classroom
you can optimize a binary classifier
tall or short
and then you can do optimal model fit to
make the the line between tall and short
you know optimize some trade-off
like recall or something like that
accuracy Model complexity you could do
three category model four category model
you can do a continuous model all these
statistical distributions and like
they're all gonna not complain they're
all gonna get optimized
so that's kind of the the blindness but
the power of
essentially just Bayesian statistics is
that once it's all lined up
it just is going to get optimized now it
may be hard to optimize it maybe easy
again it may be adequate or not
but it's kind of like within so so
just that um
yeah
it just this is what the the the

strengths and the weaknesses of
empirical modeling
which is that modeling doesn't have to
reflect
the structure of the modeled system
at all

um and it doesn't for scientists and it
doesn't for the models that are made
but some of them are adaptive still so
those are the lifelike ones but
pedagogically they don't have to be
lifelike

that's a very great example it helps and
one more question if I may and then I'll
be silent at the end of the chapter they
use the word intention you know and I
posted in the chat the the exact quote
where they say in psychological terms
this implies that preachers believe
about policies directly corresponds to
its intention

which it fulfills by acting right

I'm just a little puzzled by the use of
the word intention which typically means
motive

of an agent you know

um I don't know how it's meant here this
is where I got a little lost and what
exactly do they mean by intention here
seems an unconventional use of the trend
anyone want to give a thought on that
okay

I uh

oh yeah go for it

uh yeah this is just a speculation but
um maybe

um the meaning of the word intention
here and particularly in this paragraph

uh is related to the way it's used in um
some of some of the literature on action
philosophy so for instance
Alicia Juarez has developed a theory of
intention as a kind of
the process that results from the
causation of constraints so by intention
uh she means basically the process that
puts the system of interest in its
causal constraints so it's not
necessarily

um
maps onto or at least precisely maps
onto the psychological sense of the term
so but that's just as I said only in a
speculative thinking I may be wrong
okay another

um angle is from the paper
um active inference models do not
contradict folk psychology
it was live stream 46. does anyone who
participated in that or was there want
to just like summarize what they thought
this paper's argument was about that
issue

so basically they use the belief desire
intention framework BDI

um which I I had not heard of but it's a
commonly used framework

um and it actually came up in the first
lecture by aval in the um active
inference for the social sciences Ben
White

mentioned folks psychology and aval kind
of like followed up was like what do you
mean by like by by this basically but
the belief desire intention models just
widely used to explain

everyday behavior and intentionality
it's like what people kind of slip to
um you know this enzyme wants to bind to
this or you know prefers to bind to this
in a way just kind of broadly and they
identified those constructs belief
desire and intention
um with different constructs in the
active inference generative model
so they just basically said like
you're not wrong it just it just it's
just a description
but they uh but the beliefs are the
hidden state
beliefs
um and uh
they kind of unpack this in the context
of uh also distinguishing between
the discrete time active inference
um chapter seven and the continuous time
active inference chapter eight
and so they kind of do simulations that
pull out the
um
belief desire and intention
[Music]
pieces
um in the context of someone who's going
to the fridge
and getting ice cream
which Gene had a field day with
um yeah but but another more
deflationary way not necessarily like an
experienced intentionality like the set
point of a thermostat
or the set point of a motor
limb
you can call it what you and it's just

as long as you understand the map of
territory then such an intentionality is
is not

massively metaphysical

I found the use of the word intention
there just so like from within the
confines of the models to described as
far as chapter to just be referring to
like we're including a c uh Vector that
is preferences like those are included
along the way like there's a yeah
something like desire or some kind of
you know direction that agent is heading
as opposed to just total free form
you know going for anything right
yeah

good point about how these um intentions
or preferences or um expectations uh for
for observe observed outcomes
that those form kind of like explicit
intentions

like if you have in terms of the
directionality and the amplitude
and uh

and yeah there's a lot to say on on that
but this is what goal oriented or
um adaptive Behavior
would look like

not saying that every single
um implementation of such a system is
like you know or taking on any of that
like we don't

really talk about reward

um or goal in the active ontology

so that's why some people have very um
broad views on like intentionality and
action but the deflationary and
instrumental answer is always it's a

variational Bayesian model with action

signal processing plus control theory

Bayesian statistics

dot dot that's not even including free

energy principle or Bayesian mechanics

or going into any of that part at all

so it's purely defensible even without

fep just thought I would say that

foreign

did whoever added this question want to

explain it

um or anyone else who's here want to ask

something just while they're here

Okay so

about equation 2.6 and epistemic value

so value of new information

instead of minimizing the Divergence we

want to select policies that maximize

the expected diversions hence

Information Gain okay so only with

respect to the epistemic value

more

um informative observations is more

epistemic value

that doesn't mean that those highly

informative policies are going to have a

pragmatic value like discovering What's

um at the bottom of the ocean May um

disrupt pragmatic value about your body

temperature

um but it is an epistemic policy

how can we reconcile the above statement

with page six all facets of behavior in

cognition living organisms follow unique

imperative

minimizing the surprise and the

approximate variational free energy of

their sensory observations

anyone have a thought on this

so

models of perception and action

that do surprise minimization have the

best possible generative model

minimization of surprise is equivalent

to maximization of model evidence if you

knew exactly what to expect how much to

expect it

that's the same thing as saying you have

the best fitting model so maximization

model evidence is surprise minimization

that can't be exactly solved that's the

exact phase issue that we discussed

earlier

the most modern way to solve it is

variational accelerated optimization on

the Bayesian evidence lower bound

now you could do that about any variable

in principle

but what this imperative does is it

connects it back to sensory observations

which can be interpreted as for example

homeostatic

so minimizing surprise on homeostatic

variables

minimizing surprise via Fe

of any variable is doing the optimal

Bayesian inference on it

you could do strictly worse on

um accuracy or variance estimation but

you don't if you want to use the phase

optimal phase information Criterion

optimal model this is just how it's done

but this connects it to the homeostatic

potentially adaptive

imperative but it doesn't have to be

seen as like super adaptive and

strategic it can be seen also for like

an inert object

um

that's the variational free energy so

that's the kind of unified imperative of

real-time self-efficacy

Behavior but as was mentioned it does

not address action selection

it's a functional of beliefs cue and the

data y so we need the expected free

energy which is explicitly prospective

now within the expected free energy

there's the two terms of the that

different policies are evaluated based

on epistemic value in the pragmatic

value

such as one of the decompositions that

that I will all look forward to the um

the breakdowns on

but yes it is the case that for a given

evaluation of a policy

the more Divergent the sensory

information are

the higher the epistemic value would be

but that doesn't mean that it would be

associated with

um a pragmatic value

so an adoptive agent would

um you'd hope avoid extremes of both

extreme information seeking that risk

pragmatic value or vice versa

so that's the model fitting questioned

balancing all the parameters so that the

strategies actually adaptive

um

okay

Alexi and then we'll continue with this

question

I wanted to add to what you just said uh
Daniel just a very nice way to summarize
it that
seeking epistemic value seeking
knowledge is for humans pleasurable you
know this is an effective Neuroscience
called the seeking Drive literally the
dopaminergic exploratory system we enjoy
that this is the enthusiasm curiosity
that drives scientists and artists and
other things and when all our biological
needs are met when we're not hungry
anymore when we're safe we default back
to that system we start exploring the
environment around us to see what's out
there to gain epistemic knowledge to
maximize evidence to decrease the
surprise the negative surprise in the
future
so that we're not killed you know and
we're not dying from Hunger once we
explore the environment and find out
where's the worship and where the
Predators that helps us in the future to
avoid unpleasant surprises
nice
um
so the Divergence for the variational
free energy
let's look at two point well we we've
seen them previously
um
maximizing Divergence to maximize
Information Gain appears logical
however maximizing Information Gain
would maximize ambiguity and risk
which is
um the uncertainty about consequences of

action in the world
and it um or
um
uh this could be describing risk
um in these two lines
which are two very similar
representations of equation 2.6
um like the left term is the same
and here is observations
so this is the risk of like sort of
conditioned on a policy
um about how uncertain their uh how much
uncertainty there is about observation
sequences and here is about hidden
States
and I believe that this number is always
less than this number
because you're always more certain that
proposing something hidden
you're always more uncertain about
I believe there may be a different
reason but it it's like
you're always more uncertain about
proposed latent causes
above and beyond
um
surprise uh bounding of observables
maybe there's another way to think about
that's it
thank you Lexi wooden beliefs assign low
probability to policies
that will maximize discrepancy expected
free energy ambiguity and risk
well kind of yes and no
um
what would be the the most epistemically
salient
would be something where you were sure

about the outcomes
and they were going to contribute to
Information Gain like I have a trusty
book and I know that so I have a hundred
percent certainty about how opening the
book to this page
about kind of like the epistemic quality
so it's a Salient
um and a mapping between
the the underlying semantics and the
observation
so this representation just maybe
there's another way to see it too but or
I mean I'm sure there are but just like
this isn't like Risk like Danger
this is um
as as far as I understand
um looking at
uh
how your action
uh maps cleanly and preferably
and here's the the more sensory mapping
not to get too deep in that one point
but yeah anyone else have any
questions or thoughts on uh chapter two
Ollie that was awesome
um
work how are you gonna continue
uh well we'll continue to unpack uh all
the other at least essential equations
if not uh all the single equations in
the textbook but we're focusing on
uh developing uh similar derivations for
all the central equations in the in the
textbook and particularly from chapters
two three and uh four uh which I believe
uh constitute uh the most uh fundamental
chapters of the textbook so

um I hope we can prepare it in time so
it can be uh hopefully useful for anyone
who wants to unpack the understand them
uh a bit better

awesome

anyone else have any thoughts

well

um

we will head next into chapter three

so

why do we seem to interested only in

discrete State space discrete actions

policy discrete time is fine

yeah

um well chapter eight you're gonna love

but uh it is presented as as um there's

always a

[Music]

um

the screen continuous time are just two

options or two two families of models

that can be hybridized in chapter eight

but yeah 100 the book focuses on both

um and Thomas parr's bookstream

is uh

um

he describes the history and the

differential Focus between continuous

and discrete time models

and it's a big topic that we talk about

in chapter seven and eight

do you have any other information

yeah and even before those chapters that

we encounter uh preliminary formulation

for continuous time active imprints in

chapter four as well

yeah and five with a spinal reflex arc

so

they're just different
and uh I mean the and
the figure 4.3 the kind of Rosetta Stone
figure
is about the structural similarities
between discrete time and continuous
time models
but also understanding their differences
so chapter three is going to be the uh
High Road
so figure 1.2
chapter 2 was the base theorem so that's
why we were talking a lot about like the
fundamentals of Bayesian statistics
because up until this point this is the
same thing as just saying variational
Bays with perception and action
that is exact no someone can
implement the variational Bays with
perception and action differently but
kind of
thinking cybernetically that's what they
would need to make any kind of active
inference agents whether it's you know
using fristonian active inference or
just or not but if it has to perception
in action
chapter three is gonna take a very
different approach
that's not going to um
Focus as much on the how
as on the why
of self-organization
and the way that this is modeled with
different Bayesian architectures
this kind of prepared us on how it's
going to be done chapter 3 is a focus on
the why

and the free energy principle and then
also it's going to arrive at active
inference

chapter 4 is more about the generative
model

discrete time or continuous

chapter five is neurobiology kind of a
sampling of different kinds of
generative models how they connect
so that's kind of the next three
chapters

all right so enjoy everyone thank you
thanks again Ali and um and everyone
else who helps with the math oh yeah
there will be

um math learning groups

I believe there are Wednesdays at 13 UTC
they're on the calendar

um so uh you know whether there's one
person or more than one person

I hope people enjoy studying there and
and connecting and just improve the
documentation do whatever people feel
like doing

um but uh

there's not a specific Focus or a
facilitation around these sessions at
this point but if someone wants to
whether they're familiar with math or
not they're welcome to

um but just good luck in the math
learning group

all right

thank you

bye thanks everyone see ya